

Table of contents

Introduction

TOPIC 1: Preventive Maintenance

TOPIC 2: Conducting Maintenance

- Hardware Maintenance
- Software Maintenance

TOPIC 3: Troubleshooting

TOPIC 4: Data Protection

- Practicing Caution
- Backing Up Data

Introduction



Troubleshooting is an approach to locate the cause of faults in a computer system. Not all troubleshooting experiences are the same since technicians acquire this skill based on their knowledge and personal experience.

That is why as a computer technician, you must be **knowledgeable** in the ***fundamentals of the computer concepts*** since your approach to solving a problem may rely heavily on your own knowledge.

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TOPIC 1: Preventive Maintenance

Preventive maintenance is the regular and systematic inspection, cleaning, and replacement of worn parts, materials, and systems.

It is done to **prevent** failure of parts, materials, and systems by regularly checking whether all parts are in good working order.



Purpose of Preventive Maintenance

- Preventive maintenance reduces the probability of hardware or software problems by systematically and periodically checking hardware and software to ensure proper operation.
- Regular preventive maintenance ensures minimal system failure. When there are fewer failures, less troubleshooting will be done, thus, saving an organization time and money.
- Preventive maintenance can also be done by upgrading certain hardware or software such as a hard drive that is making a noise, upgrading insufficient memory, and installing new software updates for security and reliability.

TOPIC 2: Conducting Maintenance



First and foremost, maintenance is important in **reducing hardware and software issues**. With this, computer downtime and repair costs are reduced.

Preventive maintenance ***depends on the needs and specifications of the equipment***. For example, a device exposed to a dusty environment such as a construction site would need more maintenance as compare to an office environment.

High traffic networks such as those used in schools would **require more extensive scanning** and removal of malicious software and unwanted files.

Do take note that it is important to create documentation for any type of preventive maintenance, so that you can also check how often do certain hardware and software issues occur.

Besides these, here are more **reasons why maintenance should be conducted**:

- Increased data protection
- Extended device life
- Increased equipment stability
- Reduced repair costs
- Reduced equipment failure.

Hardware Maintenance



Maintenance for computer hardware can be done by regularly checking cables, components, and peripherals.

Regular cleaning of components should also be done to reduce overheating. Replacement of damaged components due to excessive wear should also be conducted.

The following is a **guide on conducting hardware maintenance**:

- Remove dust from fans.
- Remove dust from the power supply.
- Remove dust from components inside the computer.
- Clean the mouse and the keyboard.
- Check cables.

Software Maintenance



Software maintenance can be done by making sure that updates are current. Follow the policies when installing programs, operating system, and security updates.

Testing is done to ensure that minimal problems will be encountered when updating the software and hardware.

Here are some simple **tasks when conducting software maintenance**:

- Review and check security updates.
- Review and check software updates.
- Review and check driver updates.
- Update anti-virus files.
- Scan for viruses and malware.
- Remove unused and unwanted programs.
- Scan hard drives.
- Defragment hard drives.

TOPIC 3: Troubleshooting

Although **troubleshooting** varies and can be modified depending on case and experience, it requires an **organized and logical approach** when handling computers and components. Doing so will ensure that the process is in systematic order.

Make sure to ask the right questions, test the right hardware, and examine the correct data in order to detect the issues and resolve them.

Troubleshooting is honed over time. Every time an issue is resolved, skills on troubleshooting are also improved. Through troubleshooting experience, one would be able to know the steps needed to resolve an issue quickly.



The following is a simple process guideline which can also be modified depending on the situation:

- Explain the purpose of data protection.
- Identify the root of the problem.
- List probable causes of the issue.
- Test the theory to determine the exact cause.
- Plan the course of action to resolve the issue.
- Implement the solution.
- Check the system functionality.
- If applicable, implement preventive measures.
- Create documentation of findings, action, and outcomes.

TOPIC 4: Data Protection



Before troubleshooting, **follow the necessary precautions** in order to handle computer hardware and software properly.

Repairs such as hard drive replacement and operating system installation are **very delicate** and may **put computer data at risk**.

Make sure to prevent data loss when attempting repairs.

Practicing Caution



Data protection is one of the first steps of troubleshooting.

Computer data must **first be protected** before starting any work on a client or customer's computer.

A technician can be blamed or held liable if data loss occurs because of improper data handling.

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Backing Up Data



Backup is the copy of data on a computer or hard drive that is saved to an external media such as a DVD, CD, or flash drive.

In an organization, **back-ups are routinely done**—may it be daily, weekly, or on a monthly basis. This is to ensure that the organization's data are protected and are readily available.

If unsure whether a backup has been done, do not conduct troubleshooting activities on a client's computer.

Make sure to verify the following with the customer first:

- Date of last backup
- Content of backup
- Data integrity of the backup – Data integrity is the maintenance of the accuracy and consistency of the data. It is making sure that the data can be properly retrieved or restored in its optimum quality.
- Availability of backup media for data restoration